Sampling Identity Bridges in Symmetric Groups: Rank Recursion and Local Dynamics

Project notes from the reversible-circuits sampler experiments

September 1, 2026

Abstract

We study the following finite sampling problem. Let T_N be the set of transpositions in S_N, and let

$$\Omega(N,m) = \{(t_1,\dots,t_m) \in T_N^m : t_1 t_2 \cdots t_m = \mathrm{id}\}.$$

The target distribution is the uniform distribution on $\Omega(N,m)$. This is the random-walk bridge distribution for the random-transposition Cayley graph, conditioned to return to the identity at time m.

Our main result is a rank-recursive sampler for the dense schedule $m = 2N^3$. For every sufficiently large N, it always returns a length-$2N^3$ transposition word with product identity, runs in expected time polynomial in N and m, and has total-variation distance at most

$$3N \exp(-N/12288)$$

from the uniform bridge law. The proof combines an exact stable-sort disintegration along $S_{N-1} \subset S_N$, an exact polynomial-time sampler for prescribed-endpoint star factorizations, parity-aware pointwise smoothing, and a retained-length recursion whose errors telescope over the ranks. At the same schedule, every transposition occurs with probability at least $1 - \exp(-2N)$ under the uniform bridge. The approximation error is negligible but fixed by N and m; arbitrary requested accuracy remains open.

We also retain the local-dynamics picture. Pure Hurwitz walks are not global samplers because they preserve weighted support-component data. Exact $\{3,4,5\}$ block heat-bath updates are reversible for the uniform law and pass small exact tests, but no unconditional polynomial mixing theorem for that local chain is known. Thus rank recursion supplies the current polynomial-time sampling result, while local heat-bath mixing and bounded-degree expander gate sets remain distinct structural questions.

1 Problem Formulation

Let S_N be the symmetric group on $\{1,\dots,N\}$ and let T_N denote the conjugacy class of all transpositions. A length-m transposition word is a tuple $w = (t_1,\dots,t_m) \in T_N^m$. We say that w is an identity word if

$$\mathrm{eval}(w) = t_1 t_2 \cdots t_m = \mathrm{id}.$$

The canonical fixed-length target is

$$\Omega(N,m) = \{w \in T_N^m : \mathrm{eval}(w) = \mathrm{id}\},$$

with the uniform distribution on this finite set.

Equivalently, $\Omega(N, m)$ is the set of length-m closed walks at id in the Cayley graph $\mathrm{Cay}(S_N, T_N)$. Thus exact uniform sampling from $\Omega(N, m)$ is a random-walk bridge problem. If an exact bridge sampler were available, the transition at time j would select $t \in T_N$ with probability proportional to the number of completions from the current product times t back to id in the remaining number of steps. This is conceptually clean but not directly efficient at large N.

Throughout the mathematical statements, permutations are maps composed by

$$(\alpha\beta)(x) = \alpha(\beta(x)),$$

so the rightmost displayed factor acts first. Successive right multiplication has states

$$\mathrm{id}, \quad t_1, \quad t_1 t_2, \quad \ldots, \quad t_1 \cdots t_m,$$

and therefore evaluates to the displayed product above. Historical exact harnesses that used the opposite tuple-composition convention were translated before comparison with the statements in this note.

2 A Polynomial-Time Sampler for Dense Bridges

This section gives the main positive result. It does not analyze a Markov chain on identity words. Instead, it recursively separates transpositions that avoid the largest label from star transpositions incident to that label. Approximation enters only when long endpoint walks are replaced by their parity-uniform limits and when a negligible set of recursively short branches is discarded. Every sampled word still has exactly the prescribed length and endpoint.

For $g \in S_r$, write

$$\Omega_r(L; g) = \{(t_1, \ldots, t_L) \in T_r^L : t_1 \cdots t_L = g\}, \qquad U_{r,L,g} = \mathrm{Unif}\, \Omega_r(L; g).$$

The fiber is nonempty exactly when

$$\mathrm{sgn}(g) = (-1)^L \quad \text{and} \quad L \geq r - c(g), \tag{RR.1}$$

where $c(g)$ counts all cycles, including fixed points. Necessity follows from sign and cycle count. Sufficiency follows by writing each cycle as a minimal product of transpositions and padding by adjacent pairs.

Theorem 1 (Dense rank-recursive sampler)**.** *For every sufficiently large N and every $g \in S_N$ for which $\Omega_N(2N^3; g)$ is nonempty, there is a randomized algorithm* $\mathsf{DenseSample}(N, 2N^3, g)$ *with the following properties.*

1. *Every output consists of exactly $2N^3$ transpositions and has product exactly g.*
2. *If $\nu_{N,g}$ is its output law, then*

$$d_{\mathrm{TV}}\big(\nu_{N,g}, U_{N,2N^3,g}\big) \leq 3N \exp(-N/12288). \tag{RR.2}$$

3. *Its expected Turing runtime and space are polynomial in N and the length. A deliberately conservative bound is $O(N^4 L^3)$ exact-arithmetic operations on polynomial-bit integers and rationals, with $L = 2N^3$.*

In particular, taking $g = \mathrm{id}$ gives a polynomial-time statistically uniform sampler for dense identity bridges.

The proof is organized around four ingredients: an exact rank disintegration, exact star completion, a pointwise-smoothed one-rank lift, and a retained-length telescope.

2.1 Exact rank disintegration

Embed $H = S_{r-1}$ in S_r as the stabilizer of r. Call a transposition *old* if it lies in H, and call

$$\tau_i = (i\, r), \qquad 1 \le i < r,$$

a *star* transposition. In the group algebra put

$$C_r = \sum_{t \in T_r} t, \qquad C_{r-1} = \sum_{t \in T_{r-1}} t, \qquad J_r = \sum_{i<r} \tau_i.$$

Lemma 1 (Stable-sort bijection)**.** *Fix an old/star pattern of length L containing k star positions. There is a product-preserving bijection between words with this pattern and pairs (u, v), where u is an old word of length $L-k$ and v is a star word of length k. If $h = \mathrm{eval}(u)$ and the original endpoint is g, then*

$$\mathrm{eval}(v) = h^{-1} g. \tag{RR.3}$$

The bijection and its inverse use at most $k(L-k)$ adjacent rewrites.

Proof. For $a \in H$,

$$\tau_i a = a \tau_{a^{-1}(i)}. \tag{RR.4}$$

Repeatedly use (RR.4) to move every old factor left of every star factor. Each rewrite reduces the number of star–old inversions by one, so the procedure terminates in at most $k(L-k)$ steps and preserves the product. It is stable: no two old ancestors and no two star ancestors are exchanged.

The inverse starts from uv and restores the chosen binary pattern. Whenever a star must move left across an old factor, use the inverse equality

$$a \tau_j = \tau_{a(j)} a.$$

The two procedures undo one another rewrite by rewrite. Finally $g = \mathrm{eval}(u)\,\mathrm{eval}(v) = h\,\mathrm{eval}(v)$, which proves (RR.3). □

Since $C_r = C_{r-1} + J_r$ and C_{r-1} commutes with J_r, Lemma 1 gives the positive coefficient identity

$$[g] C_r^L = \sum_{k=0}^{L} \binom{L}{k} \sum_{h \in H} [h] C_{r-1}^{L-k}\, [h^{-1} g] J_r^k. \tag{RR.5}$$

Every summand counts actual words. Thus an exact sampler could draw (k, h) proportionally to the summand, sample the two endpoint fibers, choose a uniform binary pattern, and apply the inverse stable sort. The difficulty is to draw the labeled branch (k, h) without enumerating $(r-1)!$ possibilities.

2.2 Exact prescribed-endpoint star bridges

Write

$$B_{r,k}(q) = [q] J_r^k.$$

The following proposition turns these coefficients into an exact conditional sampler. Its enumerative input is the positive transitive star-factorization formula of Féray [2, Proposition 1.1, Theorem 1.2, and Lemma 3.1].

Proposition 1 (Star completion)**.** *Given r, k and a labeled endpoint $q \in S_r$, the number $B_{r,k}(q)$ can be computed in time polynomial in r and k. Whenever it is positive, one can sample exactly and uniformly from the corresponding length-k star words in polynomial expected time.*

Proof. Let the cycle of q containing r have length a, let $b_1, \dots, b_m$ be the lengths of the other nontrivial cycles, and let f be the number of fixed points in $\{1, \dots, r-1\}$. Put

$$\mu = (a, b_1, \dots, b_m), \quad n_0 = a + \sum_j b_j, \quad \ell_*(q) = n_0 + \ell(\mu) - 2,$$

and

$$F(x) = \frac{2\sinh(x/2)}{x} = \sum_{\substack{d \geq 0 \\ d \text{ even}}} \frac{x^d}{2^d (d+1)!}.$$

For $0 \leq z \leq f$, set

$$\mu_z = \mu \cup 1^z, \qquad n_z = n_0 + z, \qquad d_z = k - \ell_*(q) - 2z.$$

For $n_z \geq 2$, define

$$T_k(\mu_z) = \frac{k!}{n_z!} \left(\prod_{u \in \mu_z} u \right) [x^{d_z}] F(x)^{n_z - 2} \prod_{u \in \mu_z} F(ux), \tag{RR.6}$$

with a zero contribution when $d_z < 0$. Use the singleton convention $T_0((1)) = 1$ and $T_k((1)) = 0$ for $k > 0$. Decomposing a star word by the subset of fixed lower labels that it actually uses gives

$$B_{r,k}(q) = \sum_{z=0}^{f} \binom{f}{z} T_k(\mu_z). \tag{RR.7}$$

Indeed, on its exact used-label set the star factor graph is connected, so Féray's transitive formula gives (RR.6); conversely every transitive factorization on that set uses exactly those labels. This is a disjoint, positive decomposition.

All series may be truncated at degree k. There are $O(r)$ factors and $O(r)$ support choices, so schoolbook polynomial multiplication evaluates (RR.7) with polynomially many operations on polynomial-bit rationals.

For sampling, partition by the last displayed factor:

$$B_{r,j}(q) = \sum_{i<r} B_{r,j-1}(q\tau_i). \tag{RR.8}$$

Choose i with weight $B_{r,j-1}(q\tau_i)$, recurse on $(j-1, q\tau_i)$, and append τ_i. Induction using (RR.8) shows that every word has probability $1/B_{r,j}(q)$. Exact integer range draws make the procedure Las Vegas. A conservative implementation uses $O(r^3k^3)$ exact-rational operations, and all intermediate integers and rationals have polynomial bit length. □

2.3 A positive one-rank lift

For a rank-r call of length L, put $s = L - k$,

$$d = \binom{r-1}{2}, \qquad b = r - 1,$$

and define

$$A_s(h) = [h]C^s_{r-1}, \qquad B_k(q) = [q]J^k_r.$$

The exact branch weight from (RR.5) is

$$\binom{L}{k} A_s(h) B_k(h^{-1}g). \tag{RR.9}$$

We need pointwise, parity-aware estimates rather than ordinary total-variation mixing. The following form will be used twice.

Lemma 2 (Pointwise parity smoothing)**.** *Let K be the normalized random-transposition operator associated with the edges of a connected graph, acting on the corresponding symmetric group. Suppose its nontrivial absolute spectral radius, after removing the constant and sign eigenvectors, is at most ρ. On the parity class reachable at time t,*

$$\left| \frac{K^t(e,x)}{2/|G|} - 1 \right| \le \frac{|G|-2}{2}\rho^t. \tag{RR.10}$$

For the old and star walks at rank r, respectively,

$$\rho_{\mathrm{old}} = \frac{r-4}{r-2}, \qquad \rho_{\mathrm{star}} = \frac{r-2}{r-1}. \tag{RR.11}$$

Proof. Project the two point masses in the matrix coefficient $K^t(e,x)$ away from the constant and sign eigenspaces. Cauchy–Schwarz and the operator norm on the remaining space bound the absolute remainder by $(1-2/|G|)\rho^t$. Dividing by the parity-uniform mass $2/|G|$ gives (RR.10).

For an edge-transposition walk, the interchange-process spectral gap equals the one-particle graph gap by Caputo–Liggett–Richthammer [1, Theorem 1.1]. The unnormalized one-particle gaps of K_{r-1} and the star $K_{1,r-1}$ are $r-1$ and 1; division by their numbers of edges gives (RR.11). Multiplication by the sign character anticommutes with every transposition operator, so positive and negative eigenvalues are paired and the same bounds hold in absolute value. □

Consequently, on reachable endpoints,

$$\left| \frac{A_s(h)}{2d^s/(r-1)!} - 1 \right| \le \delta_{\mathrm{old}}(r,s) := \frac{(r-1)!-2}{2}\left(\frac{r-4}{r-2}\right)^s, \tag{RR.12}$$

$$\left| \frac{B_k(q)}{2b^k/r!} - 1 \right| \le \delta_{\mathrm{star}}(r,k) := \frac{r!-2}{2}\left(\frac{r-2}{r-1}\right)^k. \tag{RR.13}$$

The positive lift flattens only the longer side of (RR.9). Let $x = g^{-1}(r)$, and let $Z_r(k,x)$ be the number of length-k star words with endpoint q satisfying $q(x) = r$. These are exactly the endpoints for which $h = gq^{-1}$ fixes r. The count has the two-state recurrence

$$u_0 = 1, \quad v_0 = 0, \qquad u_{j+1} = bv_j, \quad v_{j+1} = u_j + (b-1)v_j, \tag{RR.14}$$

with $Z_r(k,r) = u_k$ and $Z_r(k,x) = v_k$ for $x \neq r$. The same completion counts give an exact backward sampler for a uniform star word satisfying the token condition.

After multiplying by one harmless common factor, use the integer k-weights

$$\widetilde{W}_k = \begin{cases} r\binom{L}{k} d^s Z_r(k,x), & s \ge k, \\ \binom{L}{k} d^s b^k, & k > s. \end{cases} \tag{RR.15}$$

If $s \geq k$, sample the star token bridge, let its endpoint be q, set $h = gq^{-1}$, and recursively request an old bridge with endpoint h. If $k > s$, sample the old word unconditionally, let its endpoint be h, and use Proposition 1 at endpoint $h^{-1}g$; this branch does not recurse. In both cases choose a uniform binary pattern and invert the stable sort.

Lemma 3 (One-rank comparison)**.** *Let*

$$\Delta(r, L) = \max\left\{\delta_{\mathrm{old}}(r, \lceil L/2 \rceil), \delta_{\mathrm{star}}(r, \lfloor L/2 \rfloor + 1)\right\}. \tag{RR.16}$$

If $\Delta(r, L) < 1$, then the exact and approximate branch supports agree, and the one-rank lift with an exact lower child differs from $U_{r,L,g}$ by at most

$$\frac{\Delta(r, L)}{1 - \Delta(r, L)} \tag{RR.17}$$

in total variation, uniformly over feasible g.

Proof. In the old-long half, (RR.12) compares every exact old endpoint count with the flat value used in (RR.15), while the star-coset weight remains exact. In the star-long half, (RR.13) compares every exact star endpoint count with the flat value, while the old-word weight remains exact. Thus every exact unnormalized branch weight divided by its approximate counterpart lies in $[1-\Delta, 1+\Delta]$. The same interval contains the ratio of normalizers. The standard likelihood-ratio estimate then gives

$$d_{\mathrm{TV}}(P, Q) \leq \frac{1}{2(1-\Delta)} \mathbb{E}_Q \left| \frac{w}{\widetilde{w}} - \frac{Z}{\widetilde{Z}} \right| \leq \frac{\Delta}{1-\Delta}.$$

Applying identical exact conditional kernels for old words, star words, patterns, and inverse sorting cannot increase total variation. □

2.4 Retained lengths and the rank telescope

One-rank smoothing is not enough: a recursive old child could be too short. The following explicit floors retain enough old length at every rank while discarding only negligible bridge mass.

Fix the top rank $N \geq 6$ and set

$$a_N = \lceil \log_2 N \rceil, \qquad z_N = \left\lceil \sqrt{N a_N} \right\rceil.$$

For $6 \leq j \leq N$, define

$$w_j = \frac{1}{j\lceil \sqrt{j} \rceil} + \frac{100 z_N}{Nj}, \qquad W_N = \sum_{j=6}^{N} w_j, \tag{RR.18}$$

$$c_5 = 1, \qquad c_r = 1 + \frac{\sum_{j=6}^{r} w_j}{W_N}, \qquad B_4 = 0, \qquad B_r = \lceil Nr(r-1)c_r \rceil. \tag{RR.19}$$

In particular $c_N = 2$ and

$$B_N = 2N^2(N-1) < 2N^3. \tag{RR.20}$$

At rank $r \geq 6$, retain every star-long branch and retain an old-long branch only if its old length s is at least B_{r-1}. Rank five needs no truncation, and ranks at most four have exact elementary samplers.

Lemma 4 (Uniform retained-rank bounds)**.** *Suppose N is large enough that*

$$\frac{100 z_N(1+\log N)}{N} \le \frac{15}{8}, \qquad 12 z_N \le N, \qquad \log N \le \frac{N}{4}. \tag{RR.21}$$

At rank five, for every $L \ge B_5$, there is no discarded branch, $\Delta(5, L) \le \exp(-5N/4)$, and every recursive child is handled exactly at rank four. Moreover, for every $6 \le r \le N$, every $L \ge B_r$, and every feasible $g \in S_r$:

1. *under $U_{r,L,g}$, the mass of discarded old-long branches is at most $\exp(-N/12288)$;*
2. *$\Delta(r, L) \le \exp(-Nr/4)$;*
3. *every retained recursive child lies in the next retained domain.*

Proof. The rank-five statement follows directly from (RR.12)–(RR.16) with $B_5 = 20N$: the two nontrivial radii are $1/3$ and $3/4$, and rank-four children are exact. We therefore turn to $r \ge 6$.

The decreasing-series and harmonic bounds, together with (RR.21), give $W_N \le 3$. In an unconditioned rank-r word, the old count A has law

$$A \sim \mathrm{Bin}\left(L, \frac{r-2}{r}\right).$$

For a fixed cutoff, its lower-tail probability decreases with L, so take $L = B_r$. Equations (RR.18)–(RR.19), including the ceiling loss, give

$$\mathbb{E}A - B_{r-1} \ge \frac{N\sqrt{r}}{32} + 6 z_N r =: t_0. \tag{RR.22}$$

Equivalently, the star count has mean at most $5Nr$. Bernstein's inequality, (RR.21), and $(x+y)^2 \ge x^2 + y^2$ yield

$$\Pr[A < B_{r-1}] \le \exp\left(-\frac{N}{12288} - 3r\log N\right). \tag{RR.23}$$

This unconditioned estimate must still be transferred to the bridge. Let $M_r = \binom{r}{2}$. Lemma 2, applied to the full transposition walk, gives on the feasible parity class

$$P_r^L(e, g) \ge \frac{2}{r!} - \left(1 - \frac{2}{r-1}\right)^L \ge \frac{1}{r!} \tag{RR.24}$$

for $L \ge B_r$ under (RR.21). Therefore $|\Omega_r(L; g)| \ge M_r^L / r!$. The number of discarded words ending at g is at most the total number of unconditioned words satisfying the tail event. Dividing by (RR.24) and using $r! \le N^r$ in (RR.23) proves

$$U_{r,L,g}(\text{discarded}) \le \exp\left(-\frac{N}{12288} - 2r\log N\right) \le \exp(-N/12288). \tag{RR.25}$$

For the second assertion, substitute $L \ge B_r$ into (RR.12)–(RR.16), use $1 - x \le e^{-x}$, and absorb the factorial prefactors using $\log r \le N/4$. This gives $\Delta(r, L) \le e^{-Nr/4}$. The third assertion is immediate from the retention rule $s \ge B_{r-1}$; star-long branches do not recurse. □

Proof of Theorem 1. At a retained call (r, L, g), compute the integer weights (RR.15), setting to zero precisely the discarded old-long weights. Draw k exactly, execute the corresponding old-long or star-long procedure, choose a uniform binary pattern, and invert Lemma 1. The endpoint is always exactly g.

Let D_r be the worst total-variation error over retained rank-r calls. Ranks at most four are exact: S_2 has one feasible word; the old parity classes at ranks three and four are singletons or exactly uniform, and the remaining all-star case uses Proposition 1. Hence $D_4 = 0$. Comparing first with the exact bridge conditioned on retention and then using Lemma 3, Lemma 4 gives

$$D_r \le \exp(-N/12288) + 2\exp(-Nr/4) + D_{r-1}. \tag{RR.26}$$

The last term follows from contraction of total variation under the exact reconstruction kernel. Summing (RR.26) from rank five to rank N gives

$$D_N \le \sum_{r=5}^{N} (\exp(-N/12288) + 2\exp(-Nr/4)) \le 3N\exp(-N/12288).$$

Equation (RR.20) places the top-level length $2N^3$ in the retained domain.

There are $O(L)$ branch weights, each with $O(L\log N)$ bits. Token sampling, permutation updates, and stable sorting are polynomial. Along one recursive path there is at most one star-long branch, hence at most one call to the $O(r^3L^3)$ star sampler. Summing over ranks gives the stated conservative $O(N^4L^3)$ exact-arithmetic bound on polynomial-bit operands. Exact fair-bit range rejection accounts for the expected-runtime qualifier. □

2.5 Complete transposition support

The dense target itself contains every transposition with overwhelming probability. This is a bridge-conditioned statement, not an ordinary coupon collector estimate.

Theorem 2 (Dense complete support)**.** *Let $M = \binom{N}{2}$, let $N \ge 3$, and let L be even. Under the uniform identity bridge,*

$$\Pr[\textit{some transposition is absent}] \le M\left(1 - \frac{1}{M}\right)^L + \frac{MN!}{2}\left(1 - \frac{2}{N}\right)^L. \tag{RR.27}$$

At $L = 2N^3$ this is at most $\exp(-2N)$. The same conclusion is immediate for $N = 2$.

Proof. Fix a transposition a. Let P be the normalized full-transposition operator and Q_a the normalized operator with a deleted. If $p_L = P^L(e,e)$ and $q_L = Q_a^L(e,e)$, exact word counting gives

$$\Pr[a \text{ absent}] = \left(\frac{M-1}{M}\right)^L \frac{q_L}{p_L}. \tag{RR.28}$$

The graph $K_N - a$ has one-particle Laplacian gap $N-2$. By [1, Theorem 1.1], sign symmetry, and the trace formula for the even return probability,

$$q_L \le \frac{2}{N!} + \left(1 - \frac{N-2}{M-1}\right)^L, \qquad p_L \ge \frac{2}{N!}.$$

Insert these estimates into (RR.28), use

$$\frac{M-1}{M}\left(1 - \frac{N-2}{M-1}\right) = 1 - \frac{2}{N},$$

and take a union bound over the M choices of a. This proves (RR.27). At $L = 2N^3$, the two terms are each at most e^{-3N} by $1 - x \le e^{-x}$, $\log M \le N$, and $\log(N!) \le N^2$. □

Corollary 1. *At length $2N^3$, the sampler of Theorem 1 fails to contain every transposition with probability at most*

$$\exp(-2N) + 3N \exp(-N/12288).$$

Proof. Apply Theorem 2 to the target and use the defining event bound for total variation. □

2.6 Scope and proof status

Theorem 1 is not an exact sampler and not an arbitrary-accuracy scheme at fixed (N, L). Its error is a negligible function of the scheduled parameters, but additional runtime does not lower the bound below (RR.2). The remaining standalone problem is to sample the positive labeled branch law (RR.9), exactly or to arbitrary requested ϵ, in time polynomial in r, L, and $\log(1/\epsilon)$ without enumerating S_{r-1} or partition-scale endpoint types.

The result is also polynomial in the permutation degree N and word length L. If S_N is used to model a reversible function on $n = \log_2 N$ wires, this is not automatically polynomial in n.

This proof is a streamlined consolidation of a multi-agent derivation. Its algebraic identities and finite boundary cases passed independent exact implementations, and the retained-length estimates were separately audited. The section is nevertheless newly assembled and should receive ordinary human referee review, with particular attention to (RR.22)–(RR.26), before being treated as a final external theorem.

3 Graph-Theoretic Interpretation

There are three useful graph-theoretic layers.

The Cayley graph layer. Let

$$X_N = \mathrm{Cay}(S_N, T_N).$$

This is the undirected $\binom{N}{2}$-regular Cayley graph generated by all transpositions. A word $w = (t_1, \dots, t_m)$ determines a walk

$$x_0 = \mathrm{id}, \qquad x_j = t_1 t_2 \cdots t_j, \qquad 0 \le j \le m,$$

in X_N. The identity-word condition is exactly $x_m = \mathrm{id}$. Thus $\Omega(N, m)$ is the set of length-m closed walks in X_N rooted at id, and the target law is the uniform closed-walk bridge measure. The intractable exact bridge sampler is the usual endpoint-conditioned random walk on X_N.

The complete-graph edge-sequence layer. Since transpositions are edges of the complete graph K_N on vertex set $[N]$, a word is also an ordered edge sequence

$$e_1, \dots, e_m \in E(K_N).$$

Evaluating the word is the token-swapping process that swaps the tokens at the endpoints of e_j at time j. Hence $\Omega(N, m)$ is the set of length-m ordered edge sequences in K_N whose token-swapping schedule returns every token to its starting vertex.

For a word w, let $G(w)$ be the support multigraph on $[N]$ whose edge multiset is $\{e_1, \dots, e_m\}$, with the time order retained separately. The identity condition is not equivalent to a simple parity

condition on $G(w)$: even edge multiplicities or even degrees are not sufficient because the swaps do not commute in general. However, $G(w)$ still records important coarse geometry. Its connected components, together with the number of time steps supported in each component, give the weighted support-component data used throughout the experiments. Disjoint support components commute, so after sorting factors with disjoint supports, the identity condition decomposes over these components.

The rewrite-graph layer. Each sampler defines a graph whose vertices are the closed schedules $\Omega(N, m)$ and whose edges are legal local rewrites. Pure Hurwitz sampling is random walk on the Hurwitz rewrite graph. Same-product block heat-bath is random walk on a larger local path-replacement graph: choose a contiguous time interval, keep the associated permutation of that window fixed, and replace the interval by another ordered edge sequence with the same endpoint in X_N.

In the complete-graph language, Hurwitz moves have a simple local form. If two adjacent edges are disjoint, the move just swaps their order. If two adjacent edges share a vertex, for example

$$(ij), (jk),$$

then a right Hurwitz move replaces them by

$$(jk), (ik).$$

Thus Hurwitz moves are generated by disjoint-edge commutations and local triangle moves in K_N. A square pair (t, t) is fixed by Hurwitz moves, whereas block heat-bath can perform square refreshes $(t, t) \mapsto (s, s)$ inside a length-4 same-product context.

With this vocabulary, the current state of the project is:

- pure Hurwitz rewrite graphs are disconnected on $\Omega(N, m)$ in small exact cases;
- the conjectural input `H1_id` says that, on one connected support component, every feasible identity schedule is connected by triangle and disjoint-commutation moves;
- the feasible identity-component lemma says that a connected component on v vertices can support an identity schedule of length k exactly when k is even and $k \geq 2(v-1)$, with doubled trees as the minimal examples;
- assuming `H1_id`, the $\{3, 4, 5\}$ same-product rewrite graph is connected for the nonexceptional even-length regimes;
- the polynomial-mixing problem is an expansion or spectral-gap problem for this local rewrite graph.

The projection/decomposition proof route is therefore a graph-expansion program: prove expansion of the projection to weighted support-component macrostates and prove expansion inside each fiber. The large-block comparison route compares the local path-replacement graph to a mesoscopic path-replacement graph. The negative canonical-routing result is also graph-theoretic: deterministic paths through a fixed doubled-tree or canonical-square normal form create superpolynomial congestion because such normal forms have exponentially small stationary mass.

4 Hurwitz Walks and Their Obstruction

For a factorization $(g_1, \ldots, g_m)$ in a group, a right Hurwitz move at position i replaces

$$(g_i, g_{i+1}) \mapsto (g_{i+1}, g_{i+1}^{-1} g_i g_{i+1}).$$

The inverse move replaces

$$(g_i, g_{i+1}) \mapsto (g_i g_{i+1} g_i^{-1}, g_i).$$

For transpositions, conjugates are transpositions, so Hurwitz moves preserve the generator set, word length, and total product.

Hurwitz moves are useful local scramblers, but they are orbit samplers rather than global samplers. The experiments track the support graph of a word: its vertices are the points touched by at least one transposition and its edges are the transpositions appearing in the word, with multiplicity. A weighted support-component invariant records the connected components of this support graph together with the number of factors supported in each component. Pure Hurwitz moves preserve this invariant in the tested cases, and the exact Hurwitz components coincide with invariant classes in the small references.

Proposition 2 (Component obstruction)**.** *Suppose a Markov chain on $\Omega(N, m)$ is confined to a Hurwitz component $C \subsetneq \Omega(N, m)$. Then, regardless of its mixing behavior inside C, its total variation distance from the global uniform law on $\Omega(N, m)$ is at least*

$$1 - \frac{|C|}{|\Omega(N, m)|}.$$

Proof. The chain assigns probability 0 to $\Omega(N, m) \setminus C$, while the uniform distribution assigns mass $1 - |C|/|\Omega(N, m)|$ to that complement. □

Exact enumeration gives the following small cases.

Case	$\lvert\Omega(N, m)\rvert$	Hurwitz components	Largest component	TV lower bound
$S_3, m = 4$	27	4	24	0.111111
$S_4, m = 4$	120	13	24	0.800000
$S_4, m = 6$	3936	17	2880	0.268293

Table 1: Exact small cases. The TV lower bound is the best possible lower bound for a pure Hurwitz walk started in the largest component.

This table is the first key negative result: pure Hurwitz random walks cannot sample the global fixed-length bridge distribution except in instances where the relevant Hurwitz orbit already has full mass.

5 Orbit-Breaking Candidates

The experiments considered two main orbit-breaking fixed-length candidates.

Block heat-bath. Choose a contiguous block $B = (t_i, \ldots, t_{i+r-1})$, compute its product $h = t_i \cdots t_{i+r-1}$, and replace B by an exactly uniform element of

$$\mathcal{B}_r(h) = \{(s_1, \ldots, s_r) \in T_N^r : s_1 \cdots s_r = h\}.$$

The outside coordinates are unchanged.

Square refresh plus Hurwitz scrambling. If an adjacent block has the form (t, t), replace it by (s, s) for a random transposition s, then apply Hurwitz scrambling. This move preserves product and length. It can break Hurwitz components, but later measurements showed that a random-slot square refresh acts too rarely at moderate N.

Pair insertion as a rank lift. A more global proposal embeds a lower-rank identity word, inserts an adjacent pair incident to the new label, applies a uniform conjugation, and then uses Hurwitz moves. This particular architecture is not exact: the unlabeled weighted support-component signature is preserved by both conjugation and Hurwitz moves, and its initial law differs from the uniform bridge law. For example, the event that all factors are one repeated edge has positive probability under the uniform target and probability zero under the one-pair lift for every $N \geq 3$ and every even length at least four. This obstruction does not apply to the rank-recursive sampler in Section 2, which samples across the positive old/star disintegration rather than retaining pair-insertion ancestry.

5.1 Stationarity of Exact Block Heat-Bath

Proposition 3 (Reversibility of block heat-bath)**.** *Fix a block-size distribution and suppose each selected block is resampled exactly uniformly from $\mathcal{B}_r(h)$, where h is the product of the old block. The resulting random-scan block heat-bath chain is reversible with respect to the uniform distribution on $\Omega(N, m)$.*

Proof. Condition on the choice of block position and size, and fix the outside coordinates. The outside coordinates force the block product h because the total word product must remain id. Under the uniform law on $\Omega(N, m)$, the conditional distribution of the selected block is precisely the uniform law on $\mathcal{B}_r(h)$. Replacing the block by an independent sample from that conditional distribution is the usual Gibbs update, hence it satisfies detailed balance. Averaging over block choices preserves reversibility. □

Consequently, if a lazy version of the chain is irreducible on $\Omega(N, m)$, it converges to the global uniform fixed-length bridge distribution. Exact conditional samplers are already available for the implemented block sizes $3, 4, 5$. The remaining local-chain tasks are therefore:

1. make the current conditional irreducibility argument unconditional;
2. prove an inverse-polynomial spectral gap, or find an asymptotic bottleneck;
3. determine whether useful larger blocks admit efficient exact conditionals and a quantitative comparison to the local chain.

6 Exact Small-Case Evidence

The first exact harness enumerated all transposition words for three smoke cases and retained the identity words. It also built the Hurwitz graph and recorded diagnostics such as prefix cycle types, edge frequencies, adjacent repeat counts, support shapes, and window product types.

The first sampler-candidate run promoted both block heat-bath and square-refresh-plus-Hurwitz on small cases: both preserved identity, visited all exact Hurwitz components in the tested cases, and had acceptable empirical diagnostic distances after finite runs. The more important result is that the exact move graphs for block heat-bath were connected in $S_3, m = 4$, $S_4, m = 4$, and $S_4, m = 6$.

Case	Total words	Identity words	Identity density
$S_3, m = 4$	81	27	0.333333333333
$S_4, m = 4$	1296	120	0.0925925925926
$S_4, m = 6$	46656	3936	0.0843621399177

Table 2: Exact enumeration harness results.

7 Finite Spectral Evidence

A later five-worker run constructed exact transition operators for block heat-bath on the small cases. The following gaps are finite-state numerical facts, not asymptotic theorems.

Case	Block sizes	States	λ_2	Gap
$S_3, m = 4$	$\{3\}$	27	0.500000	0.500000
$S_3, m = 4$	$\{4\}$	27	0.000000	1.000000
$S_3, m = 4$	$\{3, 4\}$	27	0.250000	0.750000
$S_4, m = 4$	$\{3\}$	120	0.600000	0.400000
$S_4, m = 4$	$\{4\}$	120	0.000000	1.000000
$S_4, m = 4$	$\{3, 4\}$	120	0.300000	0.700000
$S_4, m = 6$	$\{3\}$	3936	0.775573	0.224427
$S_4, m = 6$	$\{4\}$	3936	0.676284	0.323716
$S_4, m = 6$	$\{3, 4\}$	3936	0.721567	0.278433

Table 3: Exact small-case block heat-bath spectral gaps.

The same run reported that $S_5, m = 4$ and $S_5, m = 6$ each had one communication class under $\{3, 4\}$ block heat-bath moves, while Hurwitz alone had many components. These spot checks are useful evidence for irreducibility, but they do not constitute a proof.

8 Conditional Block Sampling

For a block product $h \in S_N$, define

$$F_r(h) = |\mathcal{B}_r(h)| = |\{(s_1, \ldots, s_r) \in T_N^r : s_1 \cdots s_r = h\}|.$$

Let $M = |T_N| = \binom{N}{2}$.

8.1 Prefix Rejection

The simplest exact sampler for $\mathcal{B}_r(h)$ samples the first $r - 1$ transpositions uniformly, computes the unique final factor needed to make the product equal to h, and accepts iff that final factor is a transposition. Every valid block has a unique prefix, so the accepted distribution is exactly uniform on $\mathcal{B}_r(h)$.

Lemma 5 (Prefix-rejection acceptance)**.** *The acceptance probability of prefix rejection for block size r and target product h is*

$$A_r(h) = \frac{F_r(h)}{M^{r-1}}.$$

Proof. There are M^{r-1} possible prefixes. Exactly $F_r(h)$ of them are prefixes of valid length-r factorizations of h by transpositions. □

Since T_N is a conjugacy class, $F_r(h)$ depends only on the cycle type of h. If $d(h) = N - c(h)$ is the transposition distance, where $c(h)$ is the number of cycles including fixed points, then the standard parity and distance constraints imply

$$F_r(h) = 0 \quad \text{unless} \quad d(h) \le r \quad \text{and} \quad r - d(h) \equiv 0 \pmod 2.$$

The previous run also found, with high confidence for exact counts and medium confidence for an interpolated formula table, that for fixed r,

$$A_r(h) = O_r\left(N^{2-r-d(h)}\right).$$

This asymptotic should be treated as a checked empirical/formula-guided claim until a symbolic derivation is written.

N	r	Product type	Acceptance	Expected trials
12	3	transposition	0.04959	20.17
12	3	4-cycle	0.003673	272.25
12	4	identity	0.04959	20.17
12	4	5-cycle	0.0004348	2299.97
12	5	transposition	0.004489	222.77
12	5	6-cycle	0.00006830	14641

Table 4: Selected exact prefix-rejection acceptance probabilities at $N = 12$.

Actual MCMC instrumentation confirmed the bottleneck. For $r = 4$ prefix rejection, mean attempts were about 243.6 at $N = 8, m = 20$, 1334 at $N = 10, m = 20$, 4809 at $N = 12, m = 20$, and 5282 at $N = 12, m = 30$. The largest observed attempt count in that run was 100388.

8.2 Direct Exact Samplers for $r = 4$ and $r = 5$

The direct conditional-sampler run completed the implementation target from the previous version of this note. It used the convention that permutations are stored as tuples $p[x]$, multiplication means composition "p after q", and words are evaluated left to right by right-multiplying transpositions. In this convention, after choosing first transposition t in a block with target product h, the residual target is th.

For $r = 4$, Worker A implemented an exact two-pair midpoint sampler. Let

$$P_2(g) = \{(a, b) \in T_N^2 : ab = g\}.$$

For target product h, sample a midpoint product p with probability proportional to

$$|P_2(p)|\,|P_2(p^{-1}h)|,$$

then sample uniformly from $P_2(p)$ and $P_2(p^{-1}h)$. Each valid block (t_1, t_2, t_3, t_4) has the unique midpoint $p = t_1t_2$, hence every valid block is produced with probability $1/F_4(h)$.

For $r = 5$, Worker B implemented exact completion-count recursion. Let $F_\ell(g)$ be the number of length-ℓ transposition words with product g. At residual target g, the next transposition t is chosen

Case	Blocks checked	Product count	Result
$S_3, r = 4$	81	3	pass
$S_4, r = 4$	1296	12	pass
$S_5, r = 4$	10000	60	pass
$S_3, r = 5$	243	3	pass
$S_4, r = 5$	7776	12	pass
$S_5, r = 5$	100000	60	pass

Table 5: Direct conditional block sampler validation from the completed $r = 4/r = 5$ run. "Blocks checked" means full enumeration of T_N^r for the listed case.

with probability proportional to $F_{\ell-1}(tg)$ in the project convention. The conditional probabilities telescope, so every valid length-5 block with product h has probability $1/F_5(h)$.

At $N = 20$, the $r = 4$ midpoint sampler peaked near 10.9 MB during pair-fiber construction, had worst measured target-kernel build time 21.3 ms, and had worst measured sampling time about 1.27 microseconds per sample. For a 6-cycle target at $N = 20$, direct $r = 5$ sampling took about 1.59 ms per sample, whereas prefix rejection was estimated at about 3.47 seconds per sample, a slowdown of roughly 2183×. Thus the practical prefix-rejection bottleneck for $r = 4, 5$ has been removed for the all-transpositions model.

8.3 Integrated $\{3, 4, 5\}$ Heat-Bath Kernel

The promoted kernel chooses a block size from $\{3, 4, 5\}$, chooses a contiguous block, computes its product, and replaces it by an exactly uniform block of the same length and product. Exact finite transition checks passed the standard small cases.

Case	Blocks	States	Components	Largest column error
$S_3, m = 4$	$\{3, 4, 5\}$	27	1	$5.55 \cdot 10^{-16}$
$S_4, m = 4$	$\{3, 4, 5\}$	120	1	$2.11 \cdot 10^{-15}$
$S_4, m = 6$	$\{3, 4, 5\}$	3936	1	$1.83 \cdot 10^{-14}$

Table 6: Exact finite transition checks for the integrated direct heat-bath kernel. Symmetry errors were reported as zero in these cases.

The proposed implementation interface is

`DirectTranspositionBlockSampler.count/sample_block`

together with a `block_heat_bath_step` routine. These primitives are ready to promote for the abstract model S_N with all transpositions. They do not automatically transfer to restricted reversible gate libraries, where the same-product completion counts change.

9 Scalable Diagnostics

The initial S_{10} candidate run did not enumerate $\Omega(10, m)$. It instead compared prefix cycle types to exact bridge counts from the random-transposition chain lumped by conjugacy class and measured word-level statistics.

Candidate	m	Samples	Identity failures	Max prefix TV
Block heat-bath	20	2500	0	0.1043
Square refresh + Hurwitz	20	2500	0	0.1160

Table 7: Initial S_{10} exploratory diagnostics.

Before direct block sampling, the larger scaling run tested $N \in \{8, 10, 12\}$ and $m \in \{20, 30\}$ using prefix rejection. It found zero identity failures, but prefix rejection was slow and finite chains were not close enough to justify a mixing claim: maximum prefix bridge TV was 0.2304, maximum per-position edge TV was 0.5276, and the minimum reported effective sample size was 18.38 out of 4000.

After replacing prefix rejection by direct exact conditionals, Worker D ran the direct heat-bath grid

$$N \in \{8, 10, 12, 16\}, \quad m \in \{20, 30, 40\},$$

for block sets $\{3, 4\}$, $\{4, 5\}$, and $\{3, 4, 5\}$, with burn-in 2000 and 4000 retained samples. The main summary was:

- identity failures: 0;
- block-product failures: 0;
- mean runtime per update: 0.2938 ms;
- maximum runtime per update: 0.7842 ms;
- mean option-cache hit rate: 0.5867;
- maximum prefix bridge TV proxy: 0.4843;
- maximum per-position edge TV proxy: 0.7191;
- minimum recorded effective sample size: 18.33.

The block set $\{4, 5\}$ had the best aggregate TV proxies in that run, with maximum prefix TV 0.3283 and maximum edge TV 0.6341, but it was slower than $\{3, 4\}$. The interpretation is that the exact direct kernel scales to the requested grid and preserves identity, but the convergence proxies remain too rough for any practical mixing claim.

10 Square Refresh

Square refresh is valid as a product-preserving move: (t, t) may be replaced by (s, s) because both blocks have product id. In exact small cases it helped connect Hurwitz components. However, random-slot square refresh has an action-rate bottleneck. In a mixed kernel with square-refresh probability $p = 0.05$, the measured changed-square rate was approximately 0.002055 per step for $S_{10}, m = 20$ and 0.0005479 per step for $S_{12}, m = 30$, consistent with a scale of about $p/\binom{N}{2}$. The current conclusion is to deprioritize random-slot square refresh unless a balanced square-slot proposal with detailed balance is developed.

11 Status of the Local Heat-Bath Approach

Theorem 1 gives a polynomial-time sampler for the dense schedule without proving that a local chain mixes rapidly. This section records the still-distinct status of the exact local $\{3,4,5\}$ heat-bath approach. Its stationary law and several structural inputs are known, but its polynomial mixing time remains open.

11.1 Proved inputs

The exact block heat-bath kernel is reversible for the uniform law on $\Omega(N,m)$. Therefore, for any fixed finite (N,m), a lazy irreducible version of the chain converges to the desired global uniform distribution.

Pure Hurwitz walks are confined to Hurwitz components and therefore cannot be global samplers in the small cases of Table 1.

The direct $r=4$ and $r=5$ conditional samplers are exact in the all-transpositions model. Exact block counts and per-block probabilities were validated through S_5 for both $r=4$ and $r=5$, and the integrated $\{3,4,5\}$ heat-bath transition checks passed $S_3, m=4$, $S_4, m=4$, and $S_4, m=6$.

Lemma 6 (Feasible identity components)**.** *Let C be a labeled vertex set with $|C|=v\geq 2$. There exists an identity word of length k whose factors are transpositions supported on C and whose support graph is connected if and only if*

$$k \text{ is even} \qquad \text{and} \qquad k\geq 2(v-1).$$

Proof sketch. The parity condition is forced because an identity permutation is even. The lower bound $k\geq 2(v-1)$ is the standard transitive transposition factorization lower bound specialized to the identity: a transitive factorization of a permutation $\sigma\in S_v$ has length at least $v+c(\sigma)-2$, and $c(\mathrm{id})=v$. Conversely, choose any tree on C and list its $v-1$ edges twice. The resulting doubled-tree word has product id and connected support. Additional length can be added in increments of two by inserting square pairs (t,t). □

The latest proof runs also isolated a useful macro-fiber structure. Fix the labeled weighted support-component data of a word: the nonzero support components, together with the number of factors assigned to each component. Since transpositions supported on disjoint components commute, the conditional state space factors into a color-word interleaving problem and independent transitive identity-factorization problems inside the components. The restricted heat-bath chain on such a fiber is reversible for the conditional uniform law. This factorization is now a proved structural input, but its spectral consequences remain open.

11.2 Conditional irreducibility

Earlier versions of the project isolated the following identity-specialized Hurwitz transitivity statement as the main irreducibility input. A later H1 run found a local citation to Ben-Itzhak–Teicher, Theorem 3.4, which appears to prove the needed identity Classification H statement for all-transpositions in S_N. Final manuscript use should still verify the bibliographic details and theorem numbering, but the mathematical blocker has shifted from qualitative irreducibility to quantitative mixing.

The earlier formulation is:

Conjecture 1 (`H1_id`)**.** *For every labeled set C with $|C|=v\geq 2$ and every even $k\geq 2(v-1)$, all length-k identity words in S_C whose support graph is connected lie in one Hurwitz orbit.*

Equivalently, after adding the already proved disjoint-component sorting and invariant preservation directions, `H1_id` would imply the identity Classification H: two identity transposition words are Hurwitz-connected if and only if they have the same labeled weighted support-component data.

Proposition 4 (Conditional irreducibility)**.** *Assume `H1_id`, or equivalently identity Classification H in the preceding convention. Then the lazy exact* $\{3,4,5\}$ *same-product heat-bath chain is irreducible on* $\Omega(N,m)$ *for* $N \geq 2$ *and even* $m \geq 4$*.*

Proof sketch. Size-3 same-product block updates contain padded Hurwitz moves, so the chain can move within each weighted support-component class. Size-4 updates contain square refreshes $(t,t) \mapsto (s,s)$, so the chain can change the weighted component data by creating, moving, and pruning square pairs. Under Classification H, each component can be put into a doubled-tree normal form; square refreshes then move between feasible component data, and the feasible component lemma characterizes exactly the available nonzero components. Hence all nonexceptional states communicate. □

The exceptional cases are understood separately. If m is odd then $\Omega(N,m)$ is empty. If $m = 0$ it is a singleton. If $N = 2$ and m is even it is also a singleton. If $m = 2$ and $N \geq 3$, then $\Omega(N,2) = \{(t,t) : t \in T_N\}$ is disconnected for a chain whose allowed block sizes are $3,4,5$.

11.3 Conditional polynomial-mixing statements

No unconditional polynomial mixing theorem is currently available for this local chain. The strongest output of the polynomial upper-bound attempt is a pair of conditional routes.

For any lazy reversible irreducible chain K on $\Omega(N,m)$ with spectral gap γ,

$$\tau_{\mathrm{mix}}(\epsilon) \leq \gamma^{-1}\left(m \log \binom{N}{2} + \log \frac{1}{\epsilon}\right),$$

because the uniform stationary law has $\pi_{\min} \geq |T_N|^{-m} = \binom{N}{2}^{-m}$. Thus the remaining task is an inverse-polynomial lower bound on the gap of $K_{\{3,4,5\}}$.

Theorem 3 (Projection/decomposition route, conditional)**.** *Assume `H1_id`. Let* Φ *be the labeled weighted support-component map, and let* $\bar{K}$ *be the stationary-flow projection of* $K_{\{3,4,5\}}$ *onto the image of* Φ*. Suppose the following estimates hold:*

1. *a projection Poincare inequality with constant* $q_{\mathrm{proj}}(N,m)$*;*
2. *a restricted-fiber Poincare inequality with constant* $q_{\mathrm{fib}}(N,m)$ *for every fiber of* Φ*;*
3. *a mean-lift bound with constant* $q_{\mathrm{lift}}(N,m)$ *controlling the projected Dirichlet form of fiber averages by the original Dirichlet form.*

Then

$$\mathrm{gap}(K_{\{3,4,5\}}) \geq \frac{1}{q_{\mathrm{fib}}(N,m) + q_{\mathrm{proj}}(N,m)q_{\mathrm{lift}}(N,m)}.$$

In particular, if the three constants are polynomial in N *and* m*, then* $K_{\{3,4,5\}}$ *mixes in time polynomial in* N*,* m*, and* $\log(1/\epsilon)$*.*

Proof sketch. Use the variance decomposition $\mathrm{Var}_\pi(f) = \mathrm{Var}_{\bar{\pi}}(\mathbb{E}[f \mid \Phi]) + \mathbb{E}_{\bar{\pi}}[\mathrm{Var}(f \mid \Phi)]$. Apply the projection Poincare inequality to the first term, the restricted-fiber inequality to the second term, and the mean-lift bound to compare the projected Dirichlet form back to the Dirichlet form of $K_{\{3,4,5\}}$. □

Theorem 4 (Large-block comparison route, conditional)**.** *Assume* `H1_id`*. Let K_L be the lazy ideal length-L same-product heat-bath chain. Suppose that, for some polynomially bounded $L = L(N, m)$, K_L has inverse-polynomial gap, every positive-probability K_L transition can be simulated by $\{3, 4, 5\}$ subwindow updates, and the randomized simulations have polynomial Diaconis-Saloff-Coste comparison congestion. Then $K_{\{3,4,5\}}$ has inverse-polynomial gap and hence polynomial mixing time.*

Proof sketch. Route the stationary flow of the reversible ideal chain K_L through local $\{3, 4, 5\}$ paths and apply the standard reversible-chain comparison theorem. The gap loss is the product of the inverse gap of K_L and the comparison congestion, up to absolute laziness and normalization constants. □

11.4 Current proof obligations

For the projection/decomposition route, the missing estimates are now precise. The projection must have inverse-polynomial conductance or Poincare constant; the weighted-component fibers must mix in polynomial time; and the mean-lift term must not lose more than a polynomial factor. The latest run proved a fresh-square certificate for support expansion from a length-4 identity window and found no small support-size or weighted-component bottleneck through $S_5, m = 6$. For example, the $S_5, m = 6$ support-size conductance was about 0.614126, and singleton/threshold scans of the labeled weighted-component projection had minimum conductance about 0.556800. These are finite checks, not asymptotic proofs.

For the large-block route, K_m samples the uniform bridge in one non-lazy step, but this is too global to implement locally. A useful mesoscopic K_L, with $L < m$, would require parity-aware bridge smoothing and a one-dimensional block-dynamics contraction theorem. Local simulation is false for $L = 2$ in general. For $L \geq 6$, the current reduction is to a Pair-Resampling Lemma: adjacent two-factor same-product replacements should connect every same-product length-L fiber with polynomial diameter. Finite checks through $S_5, L = 6$ found connected representative fibers, with maximum sampled-source BFS distance 12 in the largest checked fiber, but this remains empirical.

The main negative theory lesson is that deterministic routing through one canonical doubled-tree or square-rich normal form should not be used as a rapid-mixing proof. If positive comparison flow is forced through a fixed canonical word c, vertex congestion is at least on the order of $1/\pi(c)$; for a fixed canonical square word, $\pi(c) = 1/|\Omega(N, m)|$. Normal forms are therefore useful for connectivity, but a mixing proof needs randomized routes that stay in typical macrostates.

12 Alternative Gate Sets and Expander Generators

The all-transpositions model is attractive because it has rich Hurwitz structure and very explicit local block conditionals. The recent polynomial-mixing work also suggests a possible drawback: the natural $\{3, 4, 5\}$ heat-bath chain may have subtle slow modes inside weighted support-component fibers or near sparse boundary layers. It is therefore reasonable to consider different generator sets.

A particularly clean theoretical alternative is to choose, for each N, a bounded-size symmetric generating set $S_N^{\exp} \subseteq S_N$ or A_N whose Cayley graphs form an expander family. Kassabov constructed such bounded-size generating sets for symmetric and alternating groups; see arXiv:math/0503204 and arXiv:math/0505624. For the unconditioned walk, this changes the baseline geometry: the Cayley graph has bounded degree and a spectral gap bounded away from zero, so the ordinary random walk mixes in $O(\log |S_N|)$ steps up to constants.

For identity-word sampling the target would become

$$\Omega_S(N,m) = \{(g_1,\dots,g_m) \in S^m : g_1 g_2 \cdots g_m = \mathrm{id}\},$$

with the uniform law on length-m loops at the identity in $\mathrm{Cay}(G_N, S_N^{\exp})$. This is still a random-walk bridge problem, but now on a known expander rather than on the complete transposition Cayley graph equipped with a support-multigraph invariant. A natural research question is:

Can bounded-degree expander Cayley graphs support efficient near-uniform loop samplers?

The expected proof tools would be different: bridge return-ratio estimates, endpoint influence decay, and block dynamics on expander paths rather than Hurwitz orbit classification and weighted support-component geometry.

There are two caveats. First, expansion of the unconditioned Cayley walk does not by itself prove rapid mixing of a local bridge sampler. Endpoint conditioning can introduce long-range dependence, and the exact fixed-length bridge still requires either completion counts or a rapidly mixing conditioned dynamics. Second, Kassabov generators are global permutations of N points. If $N = 2^n$ is the state space of an n-wire reversible circuit, such permutations are not automatically small local reversible gates. They are best viewed as a mathematically clean benchmark model unless an efficient reversible-circuit implementation of the generators is supplied.

This suggests a useful split in future work:

- keep the all-transpositions model as the Hurwitz-rich model where exact local block samplers and support-component invariants are available;
- study expander generator sets as a benchmark model for identity bridges on bounded-degree expanding Cayley graphs;
- for cryptographic reversible circuits, compare both abstractions with implementable gate families from the random reversible-circuit literature, such as random local reversible gates or simple permutations.

If polynomial mixing for the $\{3,4,5\}$ transposition heat-bath chain is false, an expander-generator bridge model could still provide a clean theorem-level sampler target. Conversely, if even expander-generator bridge dynamics are hard to analyze, that would clarify that the main difficulty is endpoint conditioning itself rather than the transposition support geometry.

13 Implications for Reversible-Circuit Sampling

The transposition model is a clean abstraction of reversible identity-circuit sampling. In a general reversible gate library, identity circuits are closed walks in the Cayley graph of the generated permutation group. The fixed-length uniform target is again a bridge distribution. Easy samplers such as CC^{-1}, adjacent cancellations, or products of relators produce identity circuits efficiently, but they are structurally biased.

Theorem 1 shows that this obstacle can be overcome in the all-transpositions model at a dense length: one can sample a bridge negligibly close to uniform in time polynomial in the permutation degree and word length. This is a genuinely global recursive construction, not a mixing claim for Hurwitz or local heat-bath dynamics. Its use of the stabilizer chain $S_2 \subset \cdots \subset S_N$, star transpositions, and explicit symmetric-group spectra is specialized; it does not automatically transfer to an arbitrary reversible gate library.

The transposition results illustrate the discipline needed for less biased identity-circuit sampling:

- define the target distribution explicitly;
- prove the actual output law, whether by detailed balance or direct recursive disintegration;
- test exact small cases before scaling;
- track invariants that local moves may preserve accidentally;
- separate exact endpoint validity, statistical approximation, and computational pseudorandomness.

For cryptographic use, conditioning on identity means the output permutation is known. Any pseudorandomness claim must therefore concern the circuit description or the difficulty of distinguishing the generated identity word from a chosen target distribution over identity words. The present work does not prove such a cryptographic claim. It supplies a statistically natural reference sampler in one abstract gate model; converting that into a cryptographic construction still requires an efficiently represented gate library and an explicit hardness assumption about circuit descriptions.

14 Next Steps

The dense rank-recursive theorem changes the priority order. The immediate task is proof verification and simplification, followed by sharpening the one remaining approximation primitive:

1. independently referee Lemma 4 and the telescope (RR.26), including all ceiling losses, bridge-conditioning denominators, and finite-rank cases;
2. independently verify the positive star formula and its translation from Féray's convention, then compare the backward sampler against the count formula on the recorded exact cases;
3. solve the tunable positive connection problem: sample the exact branch law (RR.9), or approximate it to arbitrary requested ϵ, in time polynomial in N, L, and $\log(1/\epsilon)$;
4. improve the constants and exponent in (RR.2), which are intentionally loose and become numerically useful only at very large N;
5. determine the broadest dense length regime for which the retained-rank construction works, rather than fixing $L = 2N^3$;
6. continue the local heat-bath program as a separate structural problem: verify connected-support Hurwitz transitivity and prove or refute polynomial mixing of the exact $\{3, 4, 5\}$ chain;
7. compare the rank-recursive construction with implementable reversible gate libraries and bounded-degree expander generators before drawing cryptographic conclusions.

Internal Experimental Sources

This note consolidates the following completed local artifacts. Historical run identifiers are retained so that the underlying reports can be matched to their archived experiment directories.

- `experiments/20260617-214335-worker-a-exact-harness/report.md`;

- `experiments/20260617-214335-worker-b-hurwitz-baseline/report.md`;
- `experiments/20260617-214335-coordinator-synthesis/report.md`;
- sampler-candidates report, run `20260618-072546`;
- mixing-and-block-sampling report, run `20260618-074950`;
- direct-block-sampler-and-proof report, run `20260618-082802`;
- mixing-theory-frontier report, run `20260618-130355`;
- mixing-upper-bound-attempt report, run `20260618-160332`;
- the August 11, 2026 polynomial-mixing resolution experiment;
- the September 1, 2026 rank-recursive identity-bridge goal experiment, including its coordinator report and claim ledger;
- in particular, the stable-sort, coverage, positive-branch, exact-star, retained-closure, and red-team reports in that run's `roles/` subdirectory.

References

[1] P. Caputo, T. M. Liggett, and T. Richthammer, *Proof of Aldous' spectral gap conjecture*, Journal of the American Mathematical Society **23** (2010), 831–851. doi:10.1090/S0894-0347-10-00659-4.

[2] V. Féray, *Partial Jucys–Murphy elements and star factorizations*, European Journal of Combinatorics **33** (2012), 189–198. doi:10.1016/j.ejc.2011.09.035.